

WEEK-12

Task-1: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance

```
pavani@DESKTOP-380LEIN:~/skill$ ls -l | grep .c | sort | head
-rw-r--r-- 1 pavani pavani 65 Sep 17 15:55 combined.txt
-rw-r--r-- 1 pavani pavani 328 Sep 27 12:58 memory_test.c
-rw-r--r-- 1 pavani pavani 357 Sep 27 12:55 error_test.c
-rw-r--r-- 1 pavani pavani 360 Sep 17 13:17 append.c
-rw-r--r-- 1 pavani pavani 414 Sep 17 13:20 stderr.c
-rw-r--r-- 1 pavani pavani 542 Sep 27 12:46 process_group.c
-rw-r--r-- 1 pavani pavani 550 Sep 17 15:55 combined.c
-rw-r--r-- 1 pavani pavani 608 Sep 17 16:06 background.c
-rw-r--r-- 1 pavani pavani 642 Sep 27 12:35 signal_handlers.c
-rw-r--r-- 1 pavani pavani 650 Sep 27 12:39 sigint_shell.c
pavani@DESKTOP-380LEIN:~/skill$ cat jobtable.txt | grep Running | sort
cat: jobtable.txt: No such file or directory
pavani@DESKTOP-380LEIN:~/skill$ sleep 20 &
p 30 &
sleep 40 &[1] 1248
pavani@DESKTOP-380LEIN:~/skill$ sleep 30 &
[2] 1258
pavani@DESKTOP-380LEIN:~/skill$ sleep 40 &
[3] 1268
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]  Running                sleep 20 &
[2]-  Running                sleep 30 &
[3]+  Running                sleep 40 &
pavani@DESKTOP-380LEIN:~/skill$
```

```

pavani@DESKTOP-380LEIN:~/skill$ tops
Command 'tops' not found, did you mean:
  command 'tips' from snap tips (0.0.9)
  command 'kops' from snap kops (1.18.0-beta.1)
  command 'toms' from deb flip (1.20-7)
  command 'top' from deb procs (2:4.0.4-9ubuntu1)
  command 'stops' from deb vasttrafik-cli (1.14-1)
See 'snap info <snapname>' for additional versions.
[1] Done sleep 20
[2]- Done sleep 30
[3]+ Done sleep 40
pavani@DESKTOP-380LEIN:~/skill$ ls | invalidcommand | sort
invalidcommand: command not found
pavani@DESKTOP-380LEIN:~/skill$ time ls -l | grep .c | sort | head
-rw-r--r-- 1 pavani pavani 65 Sep 17 15:55 combined.txt
-rw-r--r-- 1 pavani pavani 328 Sep 27 12:58 memory_test.c
-rw-r--r-- 1 pavani pavani 357 Sep 27 12:55 error_test.c
-rw-r--r-- 1 pavani pavani 360 Sep 17 13:17 append.c
-rw-r--r-- 1 pavani pavani 414 Sep 17 13:20 stderr.c
-rw-r--r-- 1 pavani pavani 542 Sep 27 12:46 process_group.c
-rw-r--r-- 1 pavani pavani 550 Sep 17 15:55 combined.c
-rw-r--r-- 1 pavani pavani 608 Sep 17 16:06 background.c
-rw-r--r-- 1 pavani pavani 642 Sep 27 12:35 signal_handlers.c
-rw-r--r-- 1 pavani pavani 650 Sep 27 12:39 sigint_shell.c

real    0m0.008s
user    0m0.000s
sys     0m0.019s

```

Report: Large pipelines and multiple background jobs were executed to test the stability of the system.

CPU and memory usage were monitored using Linux process-monitoring commands.

Failed commands were also tested to check whether errors were handled without crashing the program.

Execution time was measured to study the performance of pipelines and process management.

Task-2: To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation

Architecture:



Important system calls

System Call	Purpose
fork()	Creates a child process
exec()	Executes a command
wait()/waitpid()	Waits for child processes
pipe()	Creates communication channel
dup2()	Performs I/O redirection
open()	Opens files
read()	Reads data
write()	Writes data
close()	Closes file descriptors
kill()	Sends signals

Report:

The architecture and execution flow of the project were documented using diagrams and descriptions.

The major system calls used for process creation, pipes, redirection, and signals were explained.

Important design decisions and the interaction between different modules were recorded.

The documentation was reviewed to provide a clear understanding of the complete project.

Task-3: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

Linux Shell Project

Description

A Linux shell project demonstrating process management, pipes, I/O redirection, signals, job control and inter-process communication.

Features

- Command execution
- Pipes
- Input/output redirection
- Background jobs
- Job control
- Signal handling
- Process groups
- Error handling
- Memory testing

Compilation

```
gcc program.c -o program
```

Execution

```
./program
```

System Calls

```
fork()  
exec()  
pipe()  
dup2()  
waitpid()  
open()  
read()  
write()  
close()  
kill()_
```

```
pavani@DESKTOP-380LEIN:~/skill$ find . -maxdepth 2 -type f
./data.txt
./sigtstp.c
./test_report.txt
./append.c
./jobtable
./memory_test.c
./sigint_shell
./error_test.c
./sigtstp
./jobtable.c
./process_group.c
./result.txt
./complex
./background.c
./signal_handlers.c
./README.md
./signal_handlers
./error.txt
./memory_test
./test_project.sh
./combined.c
./combined
./sigint_shell.c
./background
./stderr.c
./append
./complex.c
./combined.txt
./process_group
./stderr
./error_test
pavani@DESKTOP-380LEIN:~/skill$
```

Report: The project source code and supporting files were organized into a clear repository structure.

A README file was added to explain the project, features, compilation, and execution steps. The repository was checked to ensure that the required source files and documentation were present.

The final deliverables were reviewed and a version tag was created for the completed project.

Task-4: TO Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

```

pavani@DESKTOP-380LEIN:~/skill$ ls
README.md      combined.c      error.txt       memory_test.c   signal_handlers  test_project.sh
append         combined.txt    error_test      process_group    signal_handlers.c test_report.txt
append.c       complex         error_test.c    process_group.c  sigtstp          tests
background     complex.c      jobtable        result.txt       sigtstp.c
background.c   data.txt       jobtable.c      sigint_shell     stderr
combined       docs           memory_test     sigint_shell.c   stderr.c

pavani@DESKTOP-380LEIN:~/skill$ ls -l | grep .c
-rw-r--r-- 1 pavani pavani 360 Sep 17 13:17 append.c
-rwxr-xr-x 1 pavani pavani 16256 Sep 27 12:52 background
-rw-r--r-- 1 pavani pavani 608 Sep 17 16:06 background.c
-rwxr-xr-x 1 pavani pavani 16208 Sep 17 15:55 combined
-rw-r--r-- 1 pavani pavani 550 Sep 17 15:55 combined.c
-rw-r--r-- 1 pavani pavani 65 Sep 17 15:55 combined.txt
-rwxr-xr-x 1 pavani pavani 16336 Sep 17 16:01 complex
-rw-r--r-- 1 pavani pavani 863 Sep 17 16:01 complex.c
drwxr-xr-x 2 pavani pavani 4096 Sep 27 13:18 docs
-rw-r--r-- 1 pavani pavani 357 Sep 27 12:55 error_test.c
-rw-r--r-- 1 pavani pavani 991 Sep 17 16:16 jobtable.c
-rw-r--r-- 1 pavani pavani 328 Sep 27 12:58 memory_test.c
-rwxr-xr-x 1 pavani pavani 16304 Sep 27 12:46 process_group
-rw-r--r-- 1 pavani pavani 542 Sep 27 12:46 process_group.c
-rw-r--r-- 1 pavani pavani 650 Sep 27 12:39 sigint_shell.c
-rw-r--r-- 1 pavani pavani 642 Sep 27 12:35 signal_handlers.c
-rw-r--r-- 1 pavani pavani 814 Sep 27 12:43 sigtstp.c
-rw-r--r-- 1 pavani pavani 414 Sep 17 13:20 stderr.c
-rwxr-xr-x 1 pavani pavani 275 Sep 27 13:01 test_project.sh

pavani@DESKTOP-380LEIN:~/skill$ ls -l | grep .c | sort | head
-rw-r--r-- 1 pavani pavani 65 Sep 17 15:55 combined.txt
-rw-r--r-- 1 pavani pavani 328 Sep 27 12:58 memory_test.c
-rw-r--r-- 1 pavani pavani 357 Sep 27 12:55 error_test.c
-rw-r--r-- 1 pavani pavani 360 Sep 17 13:17 append.c
-rw-r--r-- 1 pavani pavani 414 Sep 17 13:20 stderr.c
-rw-r--r-- 1 pavani pavani 542 Sep 27 12:46 process_group.c
-rw-r--r-- 1 pavani pavani 550 Sep 17 15:55 combined.c
-rw-r--r-- 1 pavani pavani 608 Sep 17 16:06 background.c
-rw-r--r-- 1 pavani pavani 642 Sep 27 12:35 signal_handlers.c
-rw-r--r-- 1 pavani pavani 650 Sep 27 12:39 sigint_shell.c

pavani@DESKTOP-380LEIN:~/skill$ sleep 20 &
[1] 1459
pavani@DESKTOP-380LEIN:~/skill$ jobs
[1]+  Running                  sleep 20 &
pavani@DESKTOP-380LEIN:~/skill$ ./signals
-bash: ./signals: No such file or directory
[1]+  Done                     sleep 20
pavani@DESKTOP-380LEIN:~/skill$

```

Report: Final demonstration scenarios were prepared to verify the major features of the project. Pipelines, background jobs, signals, and process groups were demonstrated during the review. The complete project was checked for functionality, documentation, source organization, and reliability.

The final testing confirmed that the implemented components were ready for presentation and submission.